

helixtranslate_analysis

HelixTranslate Repository - Comprehensive Analysis Report

A. PROJECT OVERVIEW

Project Name: HelixTranslate (digital.vasic.translator) **Language:** Go 1.25.2
Module: digital.vasic.translator **Version:** 2.3.0 (VERSION file) / 3.0.0 (Makefile) **Type:** Universal multi-format, multi-language ebook translation system

Architecture Pattern: Event-driven microservices-style architecture with:

- Central event bus (pub/sub) for decoupled component communication
- Factory pattern for LLM provider instantiation
- Strategy pattern for format-specific parsers
- Hub-and-spoke pattern for WebSocket connections
- Layered architecture: cmd/ -> pkg/ -> internal/

Core Capabilities:

- Multi-format ebook parsing (FB2, EPUB, TXT, HTML, PDF, DOCX)
- Multi-language translation using 8+ LLM providers
- Real-time WebSocket monitoring dashboard
- Distributed translation via SSH workers
- REST API (HTTP/3 + WebSocket) + gRPC interfaces
- Translation caching (PostgreSQL, Redis, SQLite)
- Quality verification with multi-pass polishing
- Preparation phase analysis (characters, terminology, culture)
- Serbian Cyrillic/Latin script conversion
- Markdown workflow (EPUB <-> Markdown)

B. DIRECTORY STRUCTURE (Complete)

```
HelixTranslate/  
├── .gitignore           # Git ignore rules  
├── .gitmodules          # Submodule definitions  
(Challenges, Containers)  
├── .golangci.yml        # golangci-lint  
configuration
```

- └─ AGENTS.md # AI agent guidance (20KB)
- └─ CLAUDE.md # Claude Code guidance
- (11KB)
- └─ COMPREHENSIVE_COMPLETION_REPORT.md # Completion status report
- └─ COMPREHENSIVE_PROJECT_COMPLETION_PLAN.md # Implementation plan
- └─ CONSTITUTION.md # Team constitution
- └─ CURRENT_PROGRESS_REPORT.md # Current progress
- └─ Dockerfile # Multi-stage Docker build
- └─ DISTRIBUTED_WORK_IMPLEMENTATION_PLAN.md
- └─ DOCUMENTATION_COMPLETION_PLAN.md
- └─ Makefile # Build automation
- (v3.0.0)
- └─ README.md # WebSocket monitoring
- system docs
- └─ config.json # Main application
- configuration
- └─ go.mod # Go module dependencies
- └─ go.sum # Dependency checksums
- └─ Challenges/ # SUBMODULE - Challenge
- system
 - └─ (separate repo: vasic-digital/Challenges)
- └─ Containers/ # SUBMODULE - Container
- definitions
 - └─ (separate repo: vasic-digital/Containers)
- └─ Documentation/ # ~60 documentation files
 - └─ AGENTS.md, API.md, ARCHITECTURE.md, CLI.md
 - └─ BATCH_TRANSLATION_COMMANDS.md
 - └─ *_IMPLEMENTATION*.md, *_GUIDE*.md
 - └─ Various status/completion reports
- └─ cmd/ # 17 entry point binaries
 - └─ api-server/ # REST API server (:8080)
 - └─ challenge-runner/ # Challenge execution
- runner
 - └─ cli/ # Basic CLI tool
 - └─ deployment/ # Deployment management
 - └─ ebook-translator/ # Specialized ebook
- translator
 - └─ grpc-server/ # gRPC service (:50051)
 - └─ markdown-translator/ # Markdown workflow tool
 - └─ monitor-server/ # WebSocket monitoring
- (:8090)
 - └─ preparation-translator/ # Pre-translation analysis
 - └─ server/ # Legacy REST API server
 - └─ ssh-translation/ # SSH translation (older)
 - └─ translate-ssh/ # SSH worker binary
 - └─ translator/ # Legacy translator CLI
 - └─ unified-translator/ # PRIMARY unified CLI
- └─ internal/ # Private packages
 - └─ cache/ # Caching layer

- config/ - scripts/ - working/	# Configuration loading # 50+ shell scripts # Configs, test data, logs
pkg/	# Public packages (~30
- packages) - api/ - batch/ - coordination/ - deployment/ - distributed/ - ebook/ - events/ - fb2/	# REST handlers, routing # Batch processing # Multi-LLM coordination # Docker/SSH deployment # SSH worker system # Universal ebook parsing # Event bus system # FB2-specific XML
- handling - format/ - grpc/ - hardware/ - hash/ - language/ - logger/ - markdown/ - models/	# Format detection # gRPC service + proto # Hardware detection # Codebase hashing # Language detection # Structured logging # EPUB<->Markdown workflow # Model registry +
- downloader - preparation/ - progress/ - report/ - script/	# Pre-translation analysis # Progress tracking # Report generation # Cyrillic/Latin
- conversion - security/ - sshworker/ - storage/	# JWT auth, rate limiting # SSH worker management # DB abstraction (PG/
- Redis/SQLite) - translator/ - llm/	# Translation engine core # 8 LLM provider
- implementations - verification/ - version/ - websocket/	# Quality verification # Version management # WebSocket hub
test/	# Cross-cutting test
- suites - distributed/ - e2e/ - fixtures/ - integration/ - mocks/ - performance/ - security/ - stress/ - translator/ - unit/ - utils/	# Distributed system tests # End-to-end tests # Test data # Integration tests # Shared mocks # Performance tests # Security tests # Stress tests # Translator tests # Unit tests # Test helpers

```
└─ tests/                                     # Additional WebSocket
tests
  └─ websocket_monitoring_test.go
```

C. KEY SOURCE FILES - Detailed Analysis

C1. Entry Points

cmd/unified-translator/main.go (PRIMARY CLI)

- **Purpose:** Main unified CLI with full provider support
- **Key Types:** UnifiedConfig, TranslationSession, GeneratedFile, TranslationStep
- **Flags:** `-input/-i`, `-output/-o`, `-source-lang`, `-target-lang`, `-script`, `-provider`, `-model`, `-api-key`, `-base-url`, `-temperature`, `-max-tokens`, `-timeout`, SSH config, llama.cpp config, `-workers`, `-chunk-size`, `-concurrency`, `-verify`, `-monitoring`, `-monitoring-port`
- **Flow:** `parseFlags` -> `initLogger` -> `initEventBus` -> `createSession` -> `startMonitoring` (optional) -> `executeTranslation` -> `generateReport`
- **Providers supported:** openai, anthropic, zhipu, deepseek, qwen, gemini, ollama, llamacpp, ssh

cmd/api-server/main.go (REST API Server)

- **Purpose:** REST API server with gRPC backend connection
- **Key Types:** APIServer, APIConfig, TranslationRequest, ProviderRequestConfig, WebSocketMessage, APIResponse
- **Default ports:** gRPC :50051, HTTP :8080
- **Features:** WebSocket hub, gRPC client, JWT auth

cmd/grpc-server/main.go (gRPC Server)

- **Purpose:** gRPC translation service
- **Key Types:** ServerConfig
- **Default:** :50051 with reflection enabled
- **Flags:** `-address`, `-port`, `-max-connections`, `-reflection`, `-metrics`, `-log-level`

C2. LLM Provider Layer

pkg/translator/llm/llm.go (CORE LLM INTERFACE)

- **LLMClient interface:** `Translate(ctx, text, prompt) (string, error)`
+ `GetProviderName() string`
- **Factory:** `NewLLMTranslator(config)` -> switches on provider string
- **ValidModels map:** Maps each Provider to valid model names
- **Providers enum:** `ProviderOpenAI`, `ProviderAnthropic`, `ProviderZhipu`, `ProviderDeepSeek`, `ProviderQwen`, `ProviderGemini`, `ProviderOllama`, `ProviderLlamaCpp`, `ProviderMock`

- **Auto-retry:** `translateWithRetry()` splits text on size errors
- **Text splitting:** Splits at sentence/paragraph boundaries, max 20KB chunks
- **Size error detection:** Checks for "max_tokens", "token limit", "too large", etc.
- **Caching:** In-memory cache keyed by `text:context`

pkg/translator/llm/openai.go

- **Type:** `OpenAIClient` with `OpenAIRequest/Response` structs
- **Timeout:** 600 seconds (10 minutes for large book sections)
- **Max tokens:** 8192 (DeepSeek-compatible)
- **Default model:** gpt-4
- **Temperature:** 0.0-2.0 range validated
- **API:** POST /chat/completions

pkg/translator/llm/anthropic.go

- **Type:** `AnthropicClient`
- **API:** POST /messages with `x-api-key` and `anthropic-version: 2023-06-01`
- **Timeout:** 600 seconds
- **Max tokens:** 4096
- **Models:** claude-3-opus, claude-3-sonnet, claude-3-haiku

pkg/translator/llm/deepseek.go

- **Type:** `DeepSeekClient` (embeds `OpenAIClient`)
- **BaseURL:** <https://api.deepseek.com/v1>
- **Models:** deepseek-chat, deepseek-coder
- **Uses OpenAI-compatible API format**

pkg/translator/llm/llamacpp.go & llamacpp_provider.go

- **Local execution:** Spawns `llama.cpp` binary process
- **Configuration:** `-llama-binary`, `-llama-model`, `-context-size`
- **Models:** llama2, mistral, vicuna (custom models allowed with warning)

pkg/translator/llm/ollama.go

- **Local Ollama API:** `http://localhost:11434`
- **Models:** llama2, codellama, mistral, vicuna

pkg/translator/llm/qwen.go, zhipu.go, gemini.go

- Cloud API clients for respective providers

pkg/translator/llm/mock.go

- **MockLLMClient:** For testing, returns predictable responses

C3. Configuration System

`internal/config/config.go`

- **Config struct** with sections:
 - **ServerConfig**: Host, Port, EnableHTTP3, TLS, Timeouts, MaxUploadSize
 - **SecurityConfig**: EnableAuth, JWTSecret, APIKeyHeader, RateLimitRPS/Burst, CORSOrigins
 - **TranslationConfig**: DefaultProvider, DefaultModel, CacheEnabled, CacheTTL, MaxConcurrent, Providers map
 - **ProviderConfig**: APIKey, BaseURL, Model, Options
 - **DistributedConfig**: Enabled, Workers map, SSHTimeout, MaxRetries, HealthCheckInterval
 - **WorkerConfig**: Name, Host, Port, User, KeyFile, Password, MaxCapacity, Tags
 - **LoggingConfig**: Level, Format, OutputFile
 - **PreparationConfig**: Enabled, PassCount, Providers, Analysis flags, DetailLevel
- **Environment variable loading**:
 - OPENAI_API_KEY, ANTHROPIC_API_KEY, DEEPSEEK_API_KEY, ZHIPU_API_KEY
 - QWEN_API_KEY, GEMINI_API_KEY
 - SSH_WORKER_{HOST,USER,PASSWORD,PORT,REMOTE_DIR}
 - MONITOR_SERVER_PORT, LOG_LEVEL
- **Default config**: Port 8443, HTTP3 enabled, auth enabled, 10 RPS rate limit

C4. Translation Engine

`pkg/translator/translator.go`

- **Translator interface**: `Translate()`, `TranslateWithProgress()`, `GetStats()`, `GetName()`
- **TranslationResult**: OriginalText, TranslatedText, Provider, Cached, Error
- **TranslationStats**: Total, Translated, Cached, Errors
- **BaseTranslator**: Config, stats, in-memory cache
- **EmitProgress/EmitError**: Event bus helpers

`pkg/translator/universal.go`

- **UniversalTranslator**: Orchestrates full book translation
- **Flow**: Detect language -> Translate metadata -> Translate chapters (with progress events)
- **Per-chapter progress**: Events with chapter number, progress percentage

C5. Ebook Processing

`pkg/ebook/parser.go`

- **Book struct:** Metadata + Chapters[] + Format + Language
- **Metadata:** Title, Authors, Description, Publisher, Language, ISBN, Date, Cover
- **Chapter:** Title + Sections[]
- **Section:** Title, Content, Subsections (recursive)
- **UniversalParser:** Format auto-detection -> delegates to format-specific parser

`pkg/format/detector.go`

- **Formats:** FB2, EPUB, PDF, MOBI, AZW, AZW3, TXT, HTML, DOCX, RTF
- **Detection:** Magic bytes + file extension + content-based disambiguation
- **EPUB/DOCX disambiguation:** ZIP internal structure inspection

C6. Events & WebSocket

`pkg/events/events.go` (assumed structure)

- **EventBus:** Thread-safe pub/sub
- **Event types:** TranslationStarted, Progress, Completed, Error; ConversionStarted, Progress, Completed, Error
- **Event:** Type, Message, Data map, SessionID, Timestamp

`pkg/websocket/hub.go`

- **Hub:** Manages WebSocket clients, subscribes to event bus
- **Client:** ID, SessionID, Conn, Send channel
- **Features:** Session filtering, automatic cleanup, broadcast
- **Integration:** All events auto-forwarded to dashboard clients

C7. Distributed System

`pkg/distributed/coordinator.go`

- **DistributedCoordinator:** Manages remote LLM instances across SSH workers
- **RemoteLLMInstance:** ID, WorkerID, Provider, Model, Priority, Available
- **Discovery:** `DiscoverRemoteInstances()` queries paired workers
- **Components:** SSHPool, PairingManager, FallbackManager, VersionManager

`pkg/distributed/ssh_pool.go`

- **SSHPool:** Pooled SSH connections with cleanup
- **SSHConnection:** Wraps `*ssh.Client` with metadata
- **WorkerConfig:** SSH params, tags, capacity
- **Features:** Auto-cleanup idle connections (30 min), retry logic

C8. gRPC Layer

pkg/grpc/translator.proto

- **Service:** TranslationService with 7 RPCs
 - StartTranslation, GetTranslationStatus, ListTranslations
 - CancelTranslation, StreamTranslationProgress
 - GetProviders, SubscribeEvents
- **Messages:** TranslationRequest, ProviderConfig, TranslationOptions
 - TranslationResponse, TranslationStatusResponse
 - TranslationProgressEvent, GeneratedFile, TranslationStep
 - ProvidersResponse, ProviderInfo, ProviderStatus, SystemEvent

C9. API Handlers

pkg/api/handler.go

- **Handler struct:** Config, EventBus, Cache, AuthService, WSHub, DistributedManager
 - **Routes:**
 - GET /health, /
 - GET /ws (WebSocket)
 - POST /api/v1/translate (text translation)
 - POST /api/v1/translate/fb2 (FB2 translation)
 - POST /api/v1/translate/batch (batch translation)
 - POST /api/v1/translate/ebook (ebook translation)
 - POST /api/v1/convert/script (script conversion)
 - GET /api/v1/status/:session_id
 - GET /api/v1/version, /providers, /stats, /languages
 - POST /api/v1/translate/validate
 - POST /api/v1/translate/cancel/:session_id
 - POST /api/v1/preparation/analyze
 - GET /api/v1/preparation/result/:session_id
-

D. CONFIGURATION SYSTEM

Configuration Files Hierarchy

1. **Main config:** config.json (project root)
2. **Example configs:** internal/working/config_*.json
3. **Provider configs:** config_anthropic.json, config_deepseek.json, etc.
4. **Worker configs:** config.worker.json, config.worker.llamacpp.json
5. **Distributed configs:** config.distributed.json

Configuration Sections

```
{  
  "server": { "host", "port", "enable_http3", "tls_cert_file",  
              "tls_key_file", "timeouts", "max_upload_size" },  
}
```



```

"security": { "enable_auth", "jwt_secret", "api_key_header",
              "rate_limit_rps", "rate_limit_burst", "cors_origins" },
"translation": { "default_provider", "default_model",
                 "cache_enabled", "cache_ttl", "max_concurrent",
                 "providers": {} },
"preparation": { "enabled", "pass_count", "providers",
                 "analyze_*" flags, "detail_level" },
"distributed": { "enabled", "workers": {}, "ssh_timeout",
                 "ssh_max_retries", "health_check_interval" },
"logging": { "level", "format", "output_file" }
}

```

Environment Variables (API Keys ONLY from env)

- **OpenAI:** OPENAI_API_KEY
- **Anthropic:** ANTHROPIC_API_KEY
- **DeepSeek:** DEEPSEEK_API_KEY
- **Zhipu:** ZHIPU_API_KEY
- **Qwen:** QWEN_API_KEY
- **Gemini:** GEMINI_API_KEY
- **SSH Workers:** SSH_WORKER_HOST, SSH_WORKER_USER, SSH_WORKER_PASSWORD, SSH_WORKER_PORT, SSH_WORKER_REMOTE_DIR
- **Server:** MONITOR_SERVER_PORT, LOG_LEVEL

Security: API keys NEVER in config files (excluded by .gitignore)

- .env*, config_with_keys.json, api_keys.json, secrets.*, **/
qwen_credentials.json

E. MODEL/PROVIDER MANAGEMENT

LLM Provider Factory

NewLLMTranslator(config) -> LLMTranslator

- > validates provider string
- > validates model against ValidModels[provider]
- > creates provider-specific client via switch
- > returns LLMTranslator with embedded LLMClient

Supported Providers (9 total)

Provider	Type	API Style	Models
OpenAI	Cloud	Native	gpt-3.5-turbo, gpt-4, gpt-4-turbo, gpt-4o
Anthropic	Cloud	Native	claude-3-opus, claude-3-sonnet, claude-3-haiku
DeepSeek	Cloud	OpenAI-compatible	deepseek-chat, deepseek-coder

Provider	Type	API Style	Models
Zhipu	Cloud	Native	glm-4, glm-3-turbo
Qwen	Cloud	Native	qwen-max, qwen-plus, qwen-turbo
Gemini	Cloud	Native	gemini-pro, gemini-pro-vision
Ollama	Local	Ollama API	llama2, codellama, mistral, vicuna
LlamaCpp	Local	Process spawn	llama2, mistral, vicuna
Mock	Test	N/A	mock

Local Model Registry (pkg/models/registry.go)

Pre-configured GGUF models for local execution:

- **hunyuan-mt-7b** (Q4/Q8): Translation-specialized, 33 languages, Apache-2.0
- **aya-23-8b** (Q4): 23 languages, strong multilingual
- **qwen2.5-7b-instruct** (Q4): 32K context, Russian/Serbian support
- **mistral-7b-instruct** (Q4): General-purpose + translation
- **qwen2.5-14b-instruct** (Q4): Higher quality for 16GB+ RAM systems

Each model has: Parameters, RAM requirements, QuantType, SourceURL, Languages, Quality rating

F. EXISTING TEST STRUCTURE

Test Organization

Two locations for tests:

1. **Alongside source:** *_test.go in each pkg/* directory (standard Go pattern)
2. **Cross-cutting suites:** test/ directory

test/ Directory Structure

```
test/
├─ unit/                # 8 test files (batch, coordination, ebook,
fb2, format, language, qwen, script, verification)
├─ integration/         # 4 test files (batch_api, cross_package,
debug, ssh_translation)
├─ e2e/                 # translation_quality_e2e_test.go
├─ performance/         # translation_performance_test.go
├─ stress/              # translation_stress_test.go (.gitkeep)
├─ security/            # 3 test files (authentication,
input_validation*2)
├─ distributed/         # 6 test files (fallback, integration,
manager, performance, security, stress)
├─ translator/          # LLM provider tests
├─ mocks/               # provider.go (shared mocks)
├─ fixtures/            # Test ebooks, configs, expected
```

```
translations
└─ utils/                # helpers.go, ports.go, ssh_test_server.go,
test_infrastructure.go
```

**Test Statistics (from
COMPREHENSIVE_COMPLETION_REPORT.md)**

- **Current coverage:** 43.6% overall
- **Critical gaps:** pkg/api (32.8%), pkg/distributed (45.2%), version_manager (26.2%)
- **61 disabled tests** requiring attention
- **Build tags:** //go:build integration, //go:build e2e

Test Naming Conventions

- Table-driven tests preferred
- Naming: TestFunctionName_Scenario
- Coverage artifacts: coverage*.out, coverage.html

G. DOCUMENTATION ANALYSIS

Core Documentation Files (~60 files in Documentation/)

File	Purpose	Size
CLAUDE.md	Claude Code guidance - authoritative for Go codebase	11KB
AGENTS.md	AI agent project overview - comprehensive reference	20KB
ARCHITECTURE.md	System architecture, components, request flows	8KB
API.md	REST API documentation	15KB
API_COMPREHENSIVE_GUIDE.md	Full API guide with examples	17KB
CLI.md	CLI usage documentation	6KB
CONFIGURATION_REFERENCE.md	All config options documented	-
BATCH_TRANSLATION_COMMANDS.md	Batch translation recipes	9KB
AUTOMATED_DEPLOYMENT.md	Deployment automation	16KB
SECURITY_HARDENING.md	Security best practices	-
TESTING_GUIDE.md	Test writing guidance	-
Various *_GUIDE.md	Domain-specific guides	-
Various *_REPORT.md	Status/completion reports	-

CLAUDE.md Key Points

- Module: digital.vasic.translator, Go 1.25.2

- VERSION file (2.3.0) is authoritative over Makefile (3.0.0)
- All commands via Makefile
- Entry points in cmd/ - unified-translator is primary
- API keys from environment variables ONLY
- Distributed changes must be version-compatible
- **Definition of Done:** Real system run with pasted terminal output required

AGENTS.md Key Points

- Technology stack fully documented
 - Build/test commands documented
 - Complete directory structure
 - Code conventions (error wrapping, any over interface{}), 140 char line limit)
-

H. SUBMODULES

1. Challenges (./Challenges)

- **Repo:** git@github.com:vasic-digital/Challenges.git
- **Commit:** 3937f06e9defbc3f466e9d859cf0e34079a5fc6a
- **Purpose:** Challenge/test system - likely contains translation challenges, test suites, benchmarking tasks
- **Own submodules:** References Containers submodule
- **Files:** AGENTS.md (30KB), ARCHITECTURE.md, CLAUDE.md (27KB), CONSTITUTION.md, Makefile
- **Likely role:** Contains evaluation challenges for measuring translation quality

2. Containers (./Containers)

- **Repo:** git@github.com:vasic-digital/Containers.git
 - **Commit:** f572d2615307c696d53e56acc9fdf93f2bb6120e
 - **Purpose:** Container definitions - Dockerfiles, compose files, deployment configurations
 - **Used by:** Both main project and Challenges submodule
-

I. DEPENDENCIES

Direct Dependencies (from go.mod)

github.com/gin-gonic/gin v1.11.0	# Web framework
github.com/golang-jwt/jwt/v5 v5.3.0	# JWT authentication
github.com/google/uuid v1.6.0	# UUID generation
github.com/gorilla/websocket v1.5.3	# WebSocket support
github.com/lib/pq v1.10.9	# PostgreSQL driver
github.com/mattn/go-sqlite3 v1.14.24	# SQLite driver
github.com/quic-go/quic-go v0.56.0	# HTTP3/QUIC support
github.com/redis/go-redis/v9 v9.7.0	# Redis client

github.com/spf13/cobra v1.10.1	# CLI framework
github.com/stretchr/testify v1.11.1	# Testing framework
github.com/unidoc/unioffice v1.39.0	# DOCX processing
github.com/unidoc/unipdf/v3 v3.69.0	# PDF processing
golang.org/x/crypto v0.48.0	# TLS, SSH
golang.org/x/net v0.49.0	# Extended networking
golang.org/x/text v0.34.0	# Text processing
golang.org/x/time v0.14.0	# Rate limiting
google.golang.org/grpc v1.77.0	# gRPC framework
google.golang.org/protobuf v1.36.10	# Protocol Buffers
gopkg.in/yaml.v3 v3.0.1	# YAML parsing

Local Modules (replace directives)

- digital.vasic.challenges => ./Challenges
 - digital.vasic.containers => ./Containers
-

J. API SURFACE

REST API Endpoints (pkg/api/handler.go)

Health & Info:

- GET /health - Health check
- GET / - API info
- GET /api/v1/version - Version info
- GET /api/v1/stats - Statistics

Translation:

- POST /api/v1/translate - Text translation
- POST /api/v1/translate/fb2 - FB2 file translation
- POST /api/v1/translate/ebook - Ebook translation
- POST /api/v1/translate/batch - Batch translation
- POST /api/v1/translate/validate - Validate request
- POST /api/v1/translate/cancel/:session_id - Cancel translation

Preparation:

- POST /api/v1/preparation/analyze - Content analysis
- GET /api/v1/preparation/result/:session_id - Get analysis result

Utility:

- POST /api/v1/convert/script - Cyrillic/Latin conversion
- GET /api/v1/status/:session_id - Translation status
- GET /api/v1/providers - List providers/models
- GET /api/v1/languages - List languages

WebSocket:

- GET /ws - Real-time event streaming

gRPC Services (pkg/grpc/translator.proto)

- TranslationService.StartTranslation
- TranslationService.GetTranslationStatus
- TranslationService.ListTranslations
- TranslationService.CancelTranslation
- TranslationService.StreamTranslationProgress (server streaming)
- TranslationService.GetProviders
- TranslationService.SubscribeEvents (server streaming)

CLI Commands (cmd/unified-translator)

Full flag-based CLI with:

- Input/output file specification
 - Provider selection and configuration
 - SSH worker parameters
 - llama.cpp local parameters
 - Monitoring toggle
 - Concurrency/worker tuning
-

K. GAPS/ISSUES

Critical (from completion reports)

1. **Test Coverage Crisis:** 43.6% overall, need 100%
 - pkg/api: 32.8%
 - pkg/distributed: 45.2%
 - version_manager: 26.2%
2. **61 Disabled Tests:** Security, integration, performance suites
3. **Broken Test Infrastructure:** Hardcoded ports, race conditions, missing mocks
4. **SSH Worker Tests:** Non-existent test servers, incomplete auth mocking

Incomplete Implementations

1. **Distributed System Security:** HTTP3/QUIC pairing not implemented, SSH key management unclear
2. **Version Management:** Rollback incomplete, update verification missing
3. **Performance Management:** Dynamic scaling not implemented
4. **API Documentation:** OpenAPI spec incomplete, WebSocket events undocumented

Architecture Debt

1. **Multiple entry points:** Many cmd/ binaries with overlapping functionality

2. **Demo files at root:** Various demo-*.go and prebuilt binaries cluttering repo
3. **Coverage files committed:** coverage*.out, coverage.html in git
4. **Internal working directory:** Contains test data, logs, configs mixed together

Documentation Gaps

1. Placeholder URLs in website content
 2. Video course content missing
 3. Advanced configuration examples incomplete
 4. Distributed setup guide incomplete
-

L. INTEGRATION POINTS - Where LLMsVerifier Would Plug In

L1. LLM Provider Layer Integration (RECOMMENDED)

Location: pkg/translator/llm/ **Approach:** Add a new provider llmsverifier or create a verification wrapper

New Provider: pkg/translator/llm/llmsverifier.go

- > Implements LLMClient interface
- > Wraps any existing provider
- > Sends translations to LLMsVerifier for quality check
- > Returns verified/improved translation

Integration Points:

- llm.go: Add ProviderLLMsVerifier to Provider enum
- ValidModels: Add verifier-supported models
- Factory switch: Add case for new provider
- Config: Add LLMsVerifier section to TranslationConfig

L2. Verification Package Integration

Location: pkg/verification/ **Existing components:** verifier.go, multipass.go, polisher.go, reporter.go **Approach:** Add LLMsVerifier as a verification backend

pkg/verification/llmsverifier_backend.go

- > Implements verification backend interface
- > Integrates with existing multi-pass system
- > Can be used as additional verification pass

L3. Event-Driven Integration (Loosely Coupled)

Location: pkg/events/events.go **Approach:** Subscribe to translation completion events, trigger verification

EventBus subscriber:

```
On EventTranslationCompleted ->
    Extract translated text
    Send to LLMsVerifier API
    Emit EventTranslationVerified with results
```

L4. API Handler Integration

Location: pkg/api/handler.go **Approach:** Add verification endpoints

New endpoints:

```
POST /api/v1/verify - Submit translation for verification
GET /api/v1/verify/:id - Get verification results
POST /api/v1/translate-with-verification - Combined endpoint
```

L5. Configuration Integration

Location: internal/config/config.go **Approach:** Add LLMsVerifier configuration section

```
{
  "llmsverifier": {
    "enabled": true,
    "api_key": "${LLMSVERIFIER_API_KEY}",
    "base_url": "https://api.llmsverifier.com",
    "model": "verifier-v1",
    "verification_level": "strict",
    "auto_correct": true
  }
}
```

L6. gRPC Integration

Location: pkg/grpc/translator.proto **Approach:** Add verification RPCs

```
rpc VerifyTranslation(VerifyRequest) returns (VerifyResponse);
rpc StreamVerificationProgress(StreamRequest) returns (stream
    VerifyEvent);
```

Recommended Integration Strategy

Phase 1: Add LLMsVerifier as a new LLM provider in pkg/translator/llm/ - minimal changes, uses existing factory pattern **Phase 2:** Integrate with pkg/verification/ for multi-pass quality checks **Phase 3:** Add REST/gRPC API endpoints for standalone verification **Phase 4:** Add configuration and event-driven integration

Summary Statistics

- **Total files in repo:** ~936
- **Go source packages:** ~30 in pkg/ + internal/
- **Entry point binaries:** 17 in cmd/
- **LLM providers:** 9 (8 real + 1 mock)
- **Test files:** 100+ across test/ and pkg/*_test.go
- **Documentation files:** 60+ in Documentation/ + root
- **Supported formats:** 10 (FB2, EPUB, TXT, HTML, PDF, DOCX, MOBI, AZW, AZW3, RTF)
- **Submodules:** 2 (Challenges, Containers)
- **Direct Go dependencies:** 19 + 2 local replace